

Analysis group meeting

Mathieu Ouillon (Mississippi State University)

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- We had issues with the CND, RICH, and helicity timelines.
- Chris investigates the log to understand.
- He retries the first step (production of the histograms (usually done by the workflow)) locally (on the interactive farm).
- He has no issues, he run the second step to produce the timelines.
- No errors again.
- He still doesn't understand the issues I faced before. He still works on that.
- But now we have the timelines for the CND, RICH and helicity.

- About the timeline forward electron V_z :
- I talk to Chris, and he asks me to make a pull request with the fit function.
- It is done with a double Gaussian and a first-order polynomial.
- I need to add code to apply this new fit function only to RG-D runs.

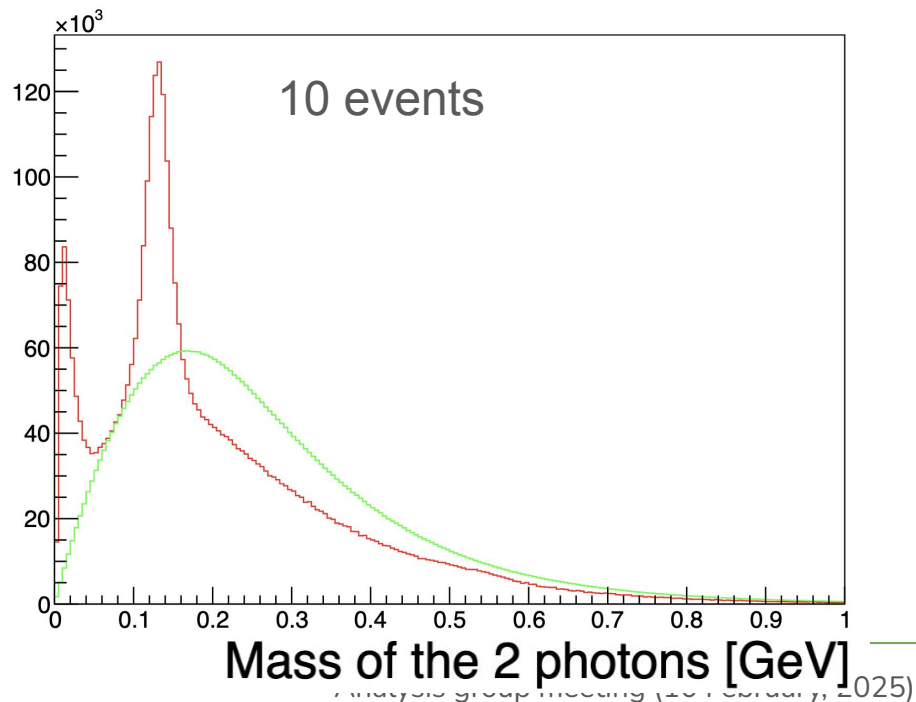
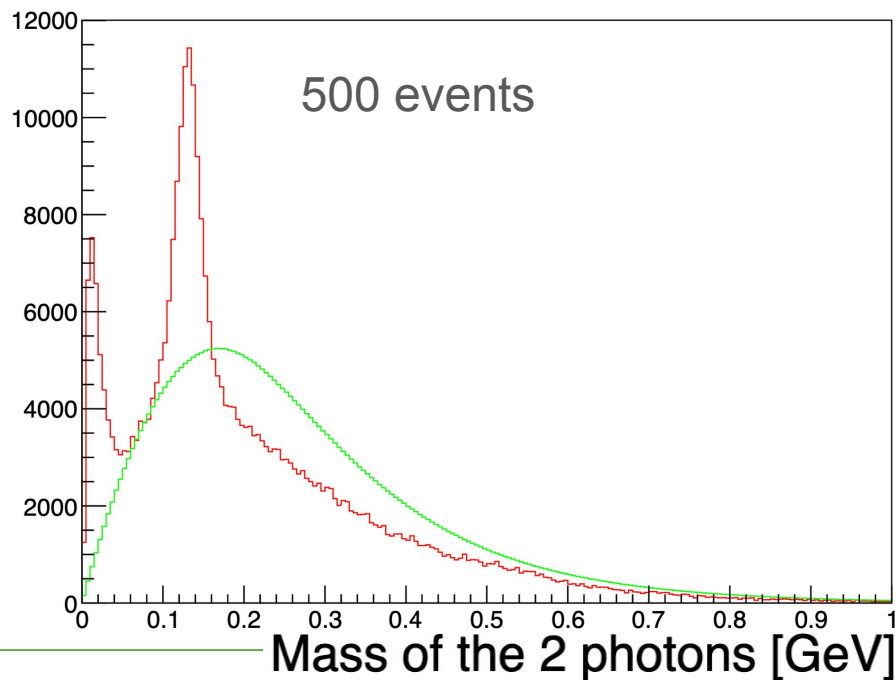
RG-D study: π^0 analysis

4

- Data: pass0v7 LD2 data
- Cuts: Electron pid=11, status<0 and $-11 < v_z < 0$ cm, $\theta_{e\gamma} > 8$ deg.
- Photons: pid =22, $E > 0.2$ GeV and $-1.67 < \Delta t < 1.55$ ns

Red - main events

Green - event mixing



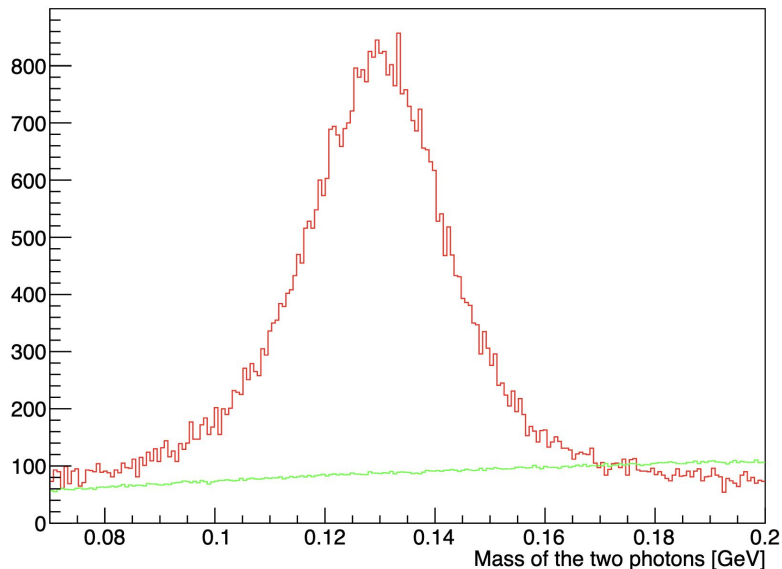
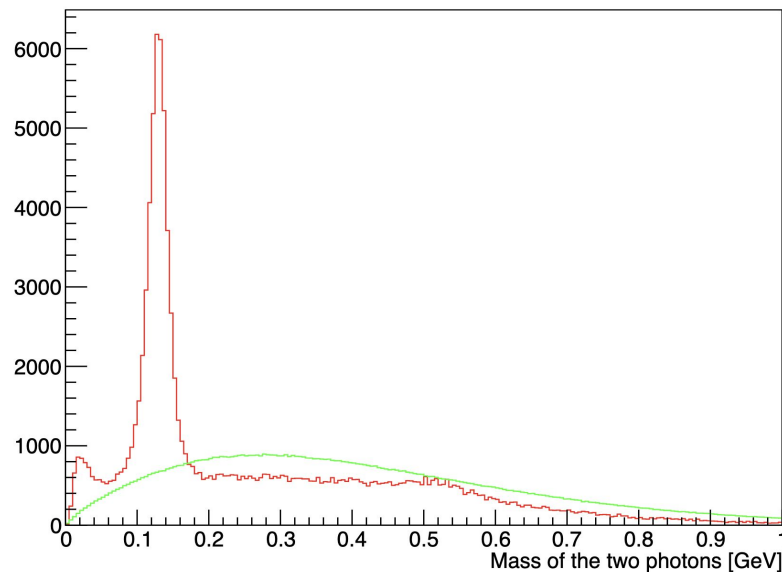
RG-D study: π^0 analysis

- Data: pass0v7 LD2 data
- Cuts: Electron pid=11, status<0 and $-11 < v_z < 0$ cm
- Photons: pid =22, $E > 0.2$ GeV and $-1.67 < \Delta t < 1.55$ ns, $\theta_{ey} > 8$ deg.
- DIS cuts: $Q^2 > 1$ GeV², $W > 2$ GeV, $0.3 < z_h < 0.7$

Red - main events

Green - event mixing

10 combinations



RG-D Nuclear transparency

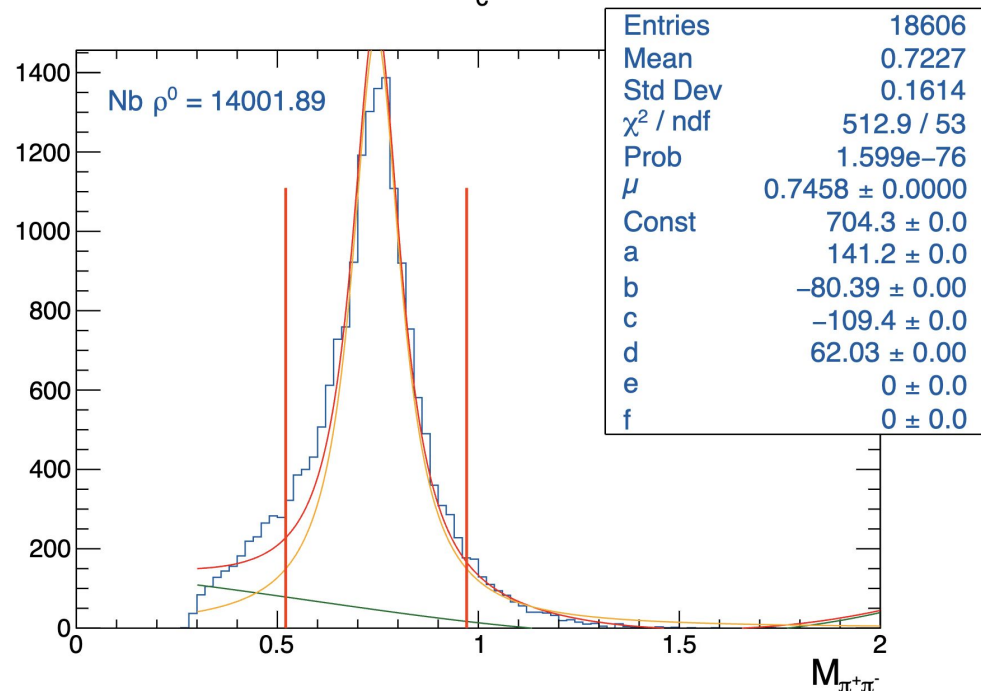
- Use Pass0v7, all outbending data.
- Electron cuts: pid=11, status<0, $-5 < \chi^2_{\text{pid}} < 5$, $-20 < V_z < 5$ cm (LD2), $-10.53 < V_z < 5$ cm (Cx C), $-10.50 < V_z < -6.45$ cm (Cu), $-6.00 < V_z < 5$ cm (Sn)
- Pi plus cuts: pid=211, $-10 < \chi^2_{\text{pid}} < 10$, $-20 < V_z < 5$ cm (LD2), $-9.65 < V_z < 5$ cm (Cx C), $-9.76 < V_z < -5.16$ cm (Cu), $-4.64 < V_z < 5$ cm (Sn)
- Pi minus cuts: pid=-211, $-10 < \chi^2_{\text{pid}} < 10$, $-20 < V_z < 5$ cm (LD2), $-10.35 < V_z < 5$ cm (Cx C), $-10.95 < V_z < -5.83$ cm (Cu), $-5.46 < V_z < 5$ cm (Sn)
- Kinematics cuts: $Q^2 > 2 \text{ GeV}^2$, $0.9 < z_h < 1$, $0.1 < t < 0.5 \text{ GeV}^2$, $l_c < 1 \text{ fm}$ (l_c apply only for T_A vs Q^2)
- Binning in Q^2 : {1.0, 2.0, 2.5, 3.0, 3.5, 4.5, 6.0}, To extract p^0 we use this function for each bin and integrating the Breit Wigner between $\mu - 0.15$ and $\mu + 0.15$.

$$f(x; \alpha, \mu, p_1, p_2, p_3) = \alpha \frac{1}{2\pi} \frac{\Gamma/2}{(x - \mu)^2 + (\Gamma/2)^2} + p_0 + p_1 x + p_2 x^2 + p_3 x^3,$$

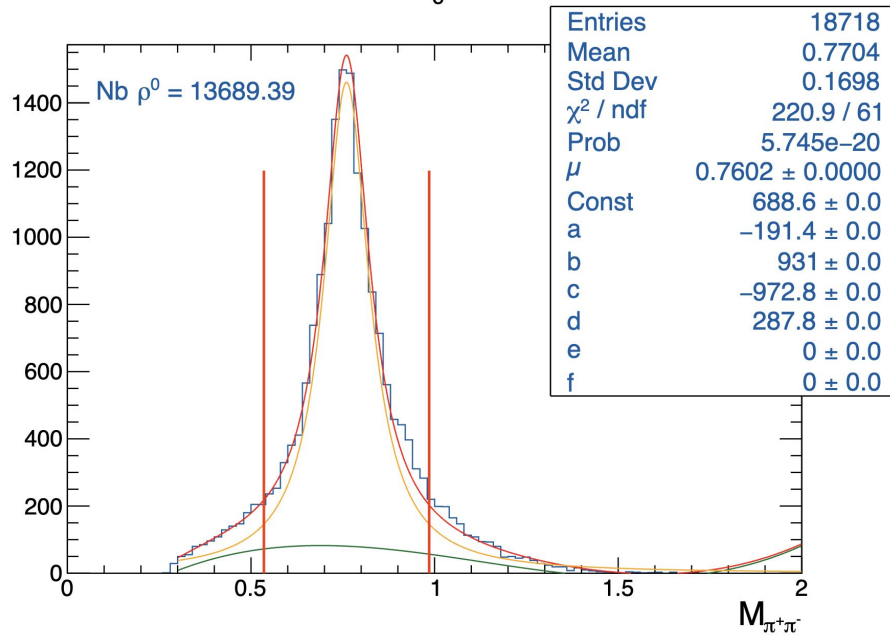
$$\Gamma/2 = 0.150 \text{ GeV}$$

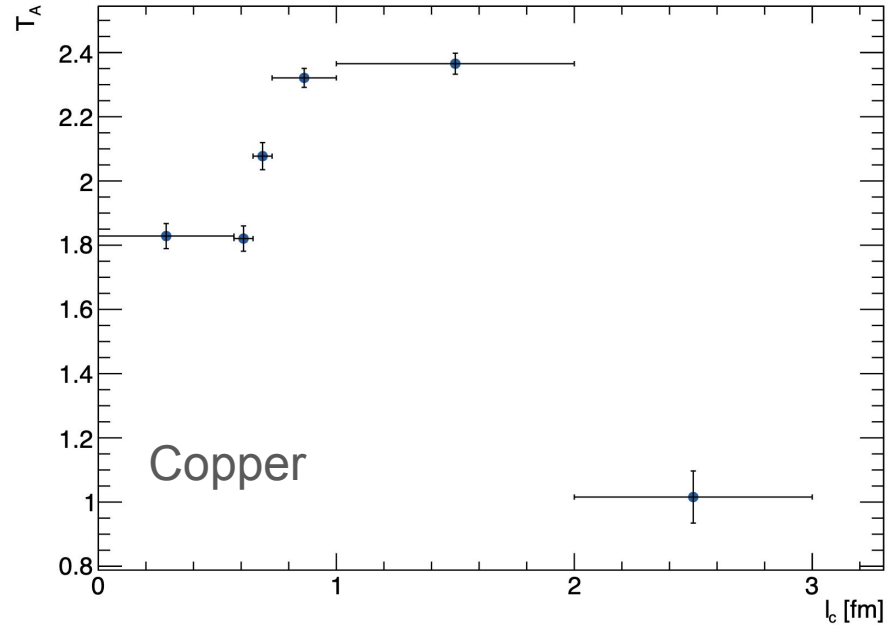
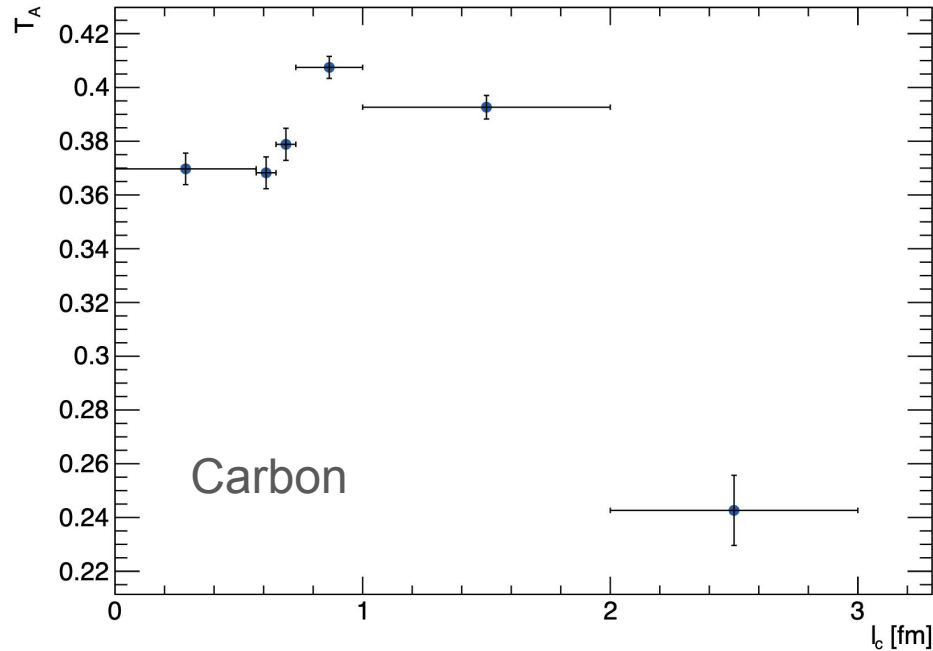
LD2

$0.00 < l_c < 0.57$

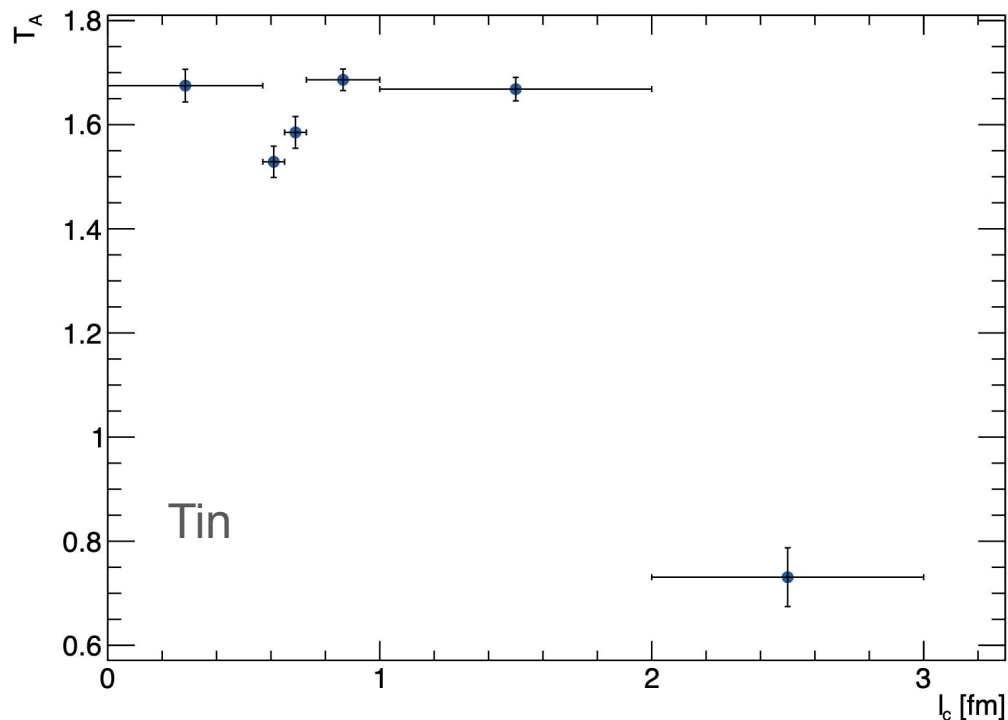


$0.57 < l_c < 0.65$





RG-D Nuclear transparency as function of l_c



Nuclear transparency:

$$T_A = \frac{N_A^{\rho^0}}{N_{LD2}^{\rho^0}} \frac{l_{LD2} \rho_{LD2}}{l_A \rho_A}$$

$$\sigma_{T_A} = T_A \sqrt{\frac{1}{N_A^{\rho^0}} + \frac{1}{N_{LD2}^{\rho^0}}}$$

with $A : \{C, Cu, Sn\}$

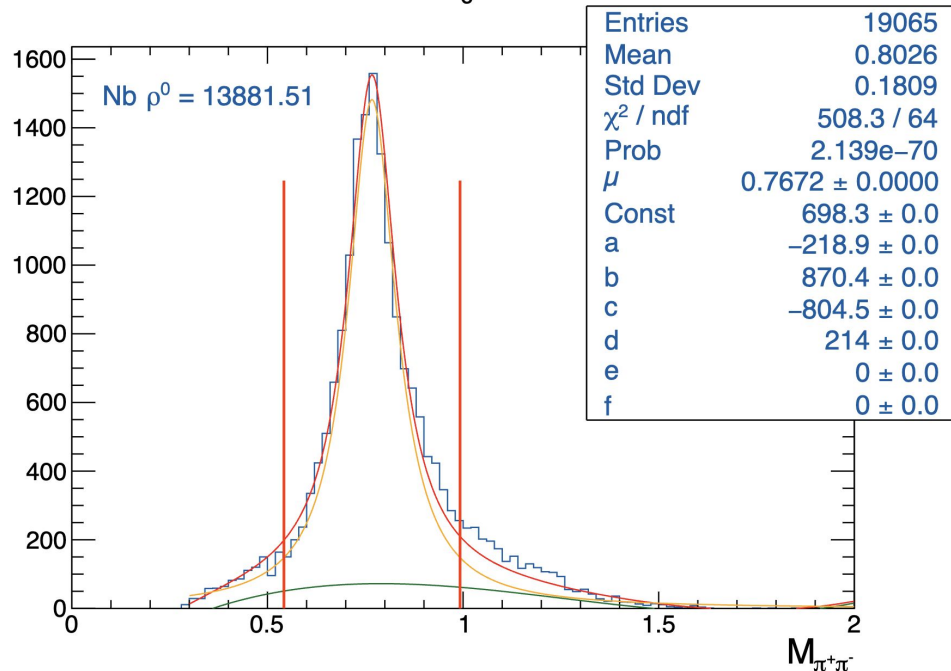
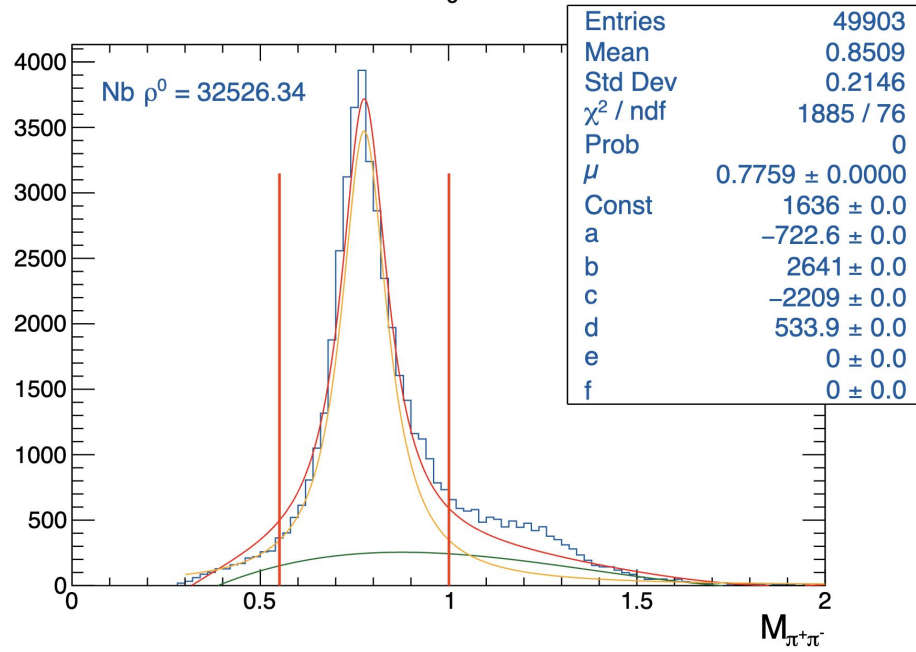
with $l_{LD2} = 5$ cm, $l_C = 0.4$ cm, $l_{Cu} = 0.0093$ cm or $l_{Sn} = 0.0171$ cm

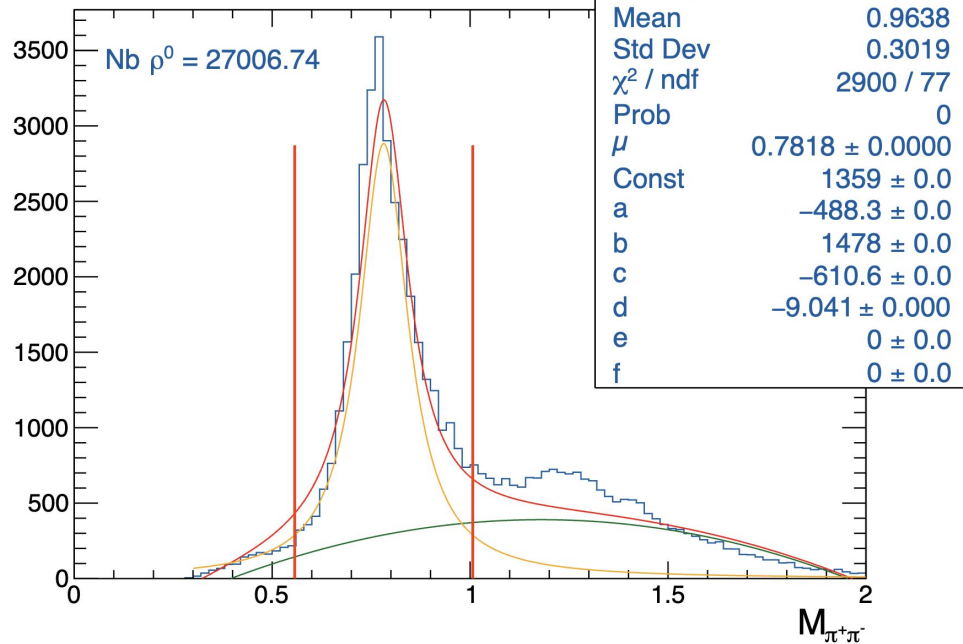
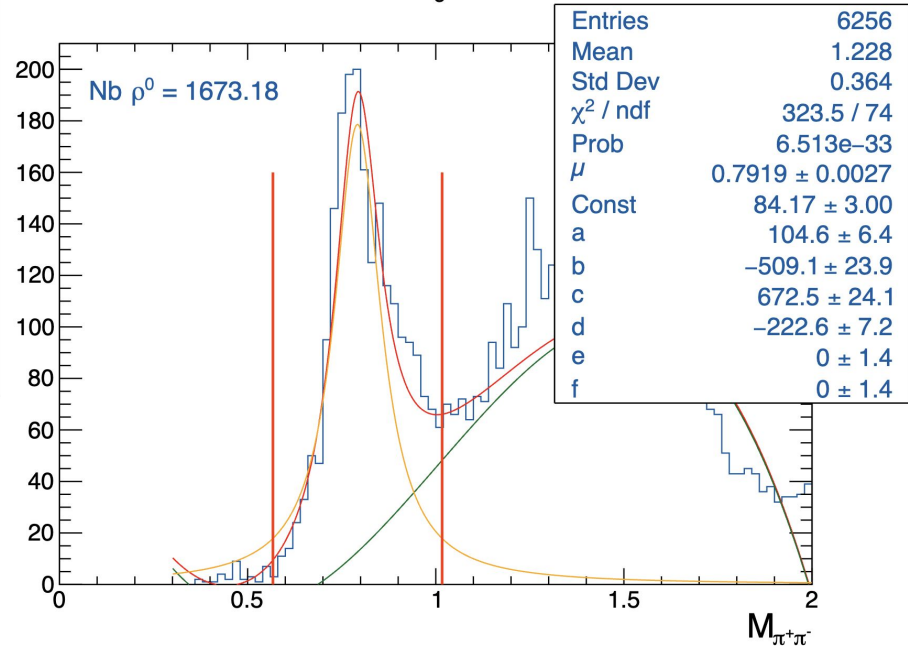
$\rho_{LD2} = 0.164$ g/cm³, $\rho_C = 2.2$ g/cm³,

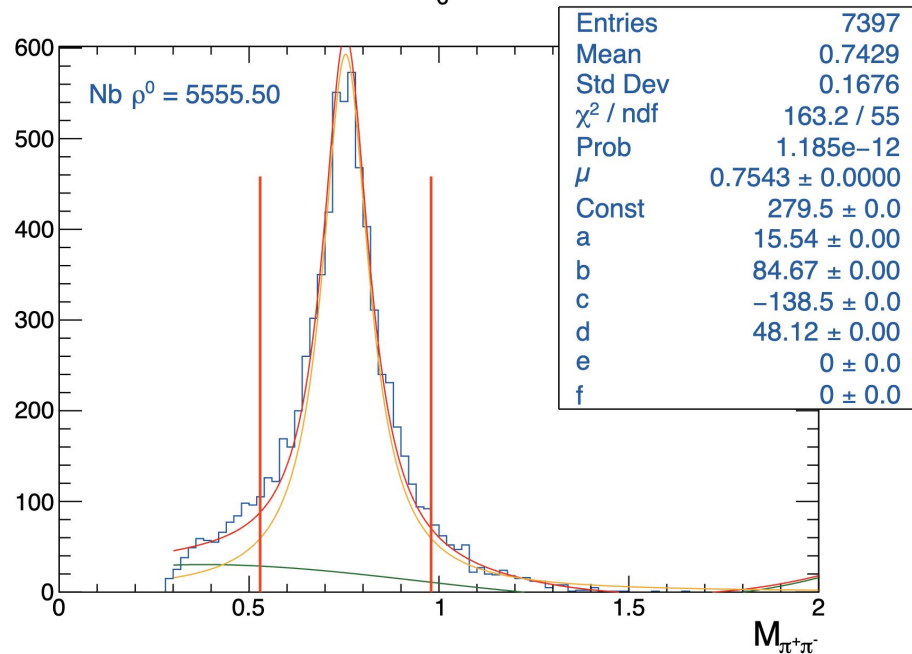
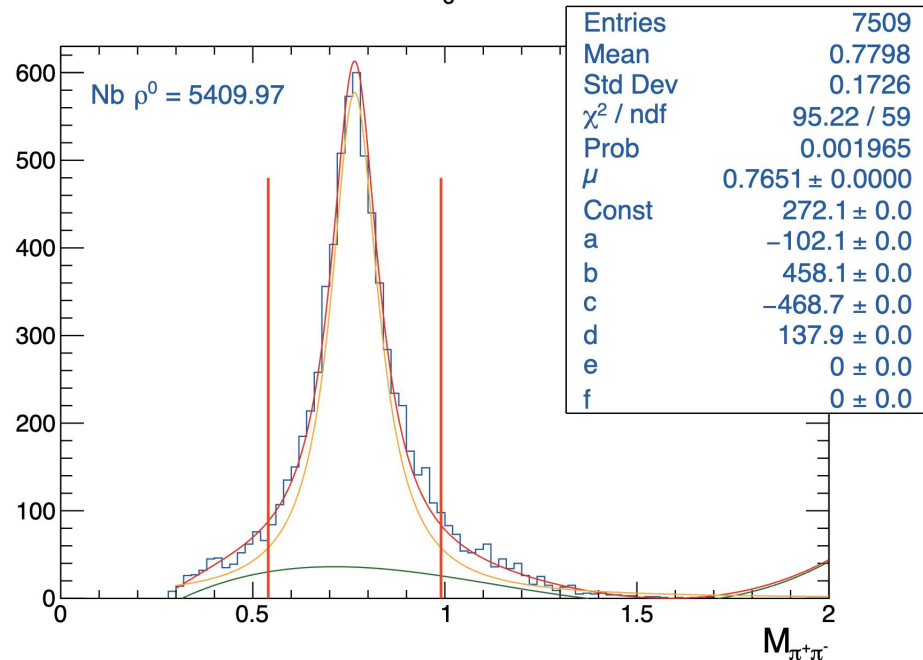
$\rho_{Cu} = 8.96$ g/cm³, $\rho_{Sn} = 7.31$ g/cm³

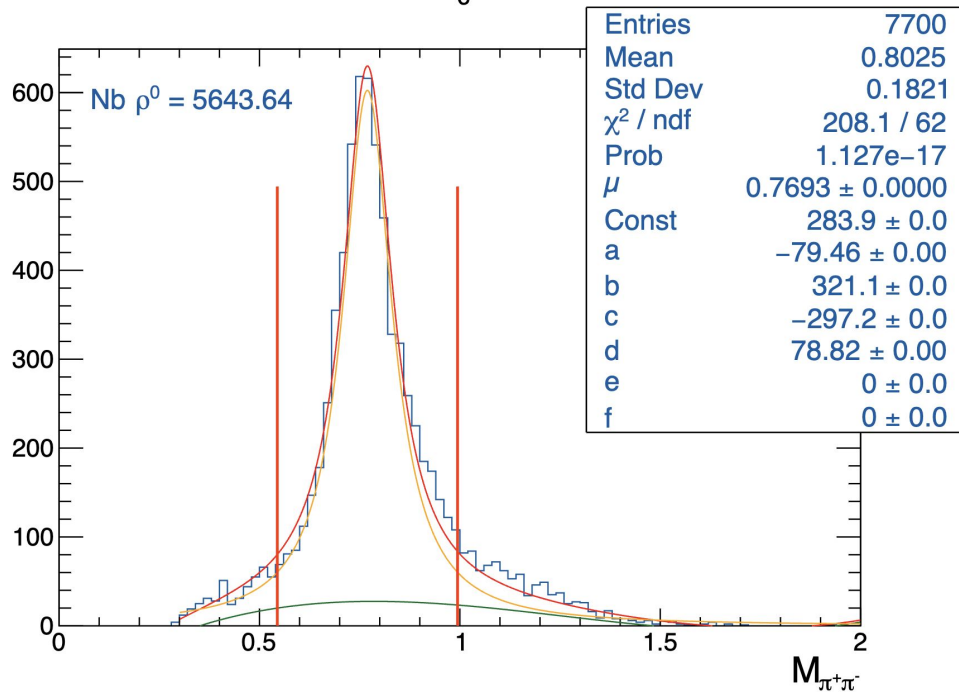
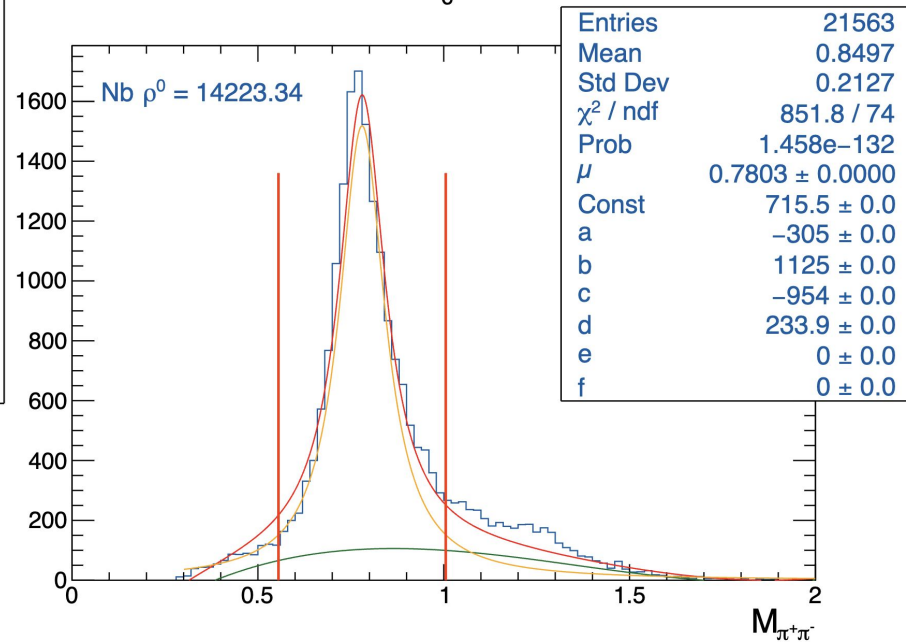
- RG-D analysis:
 - Cross check for the nuclear transparency vs Q^2
 - Work on π^0 analysis:
 - Fix the event mixing method
- Background merging, generate 100k events for one configuration
- AI for ALERT:
 - Redo the efficiency/purity analysis using luminosity simulation to have a better background
 - Comparison between conventional and AI track finding

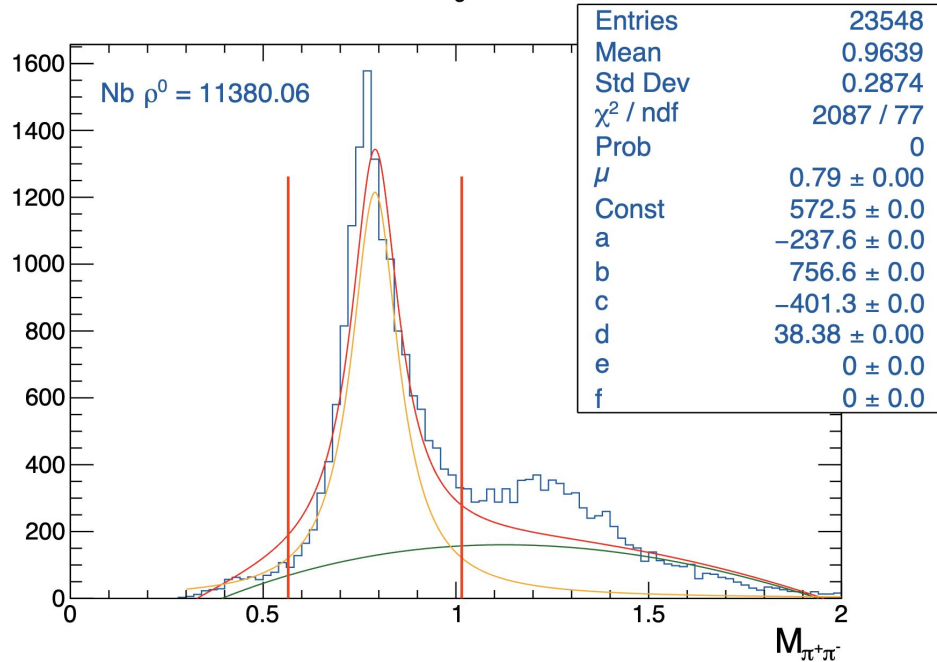
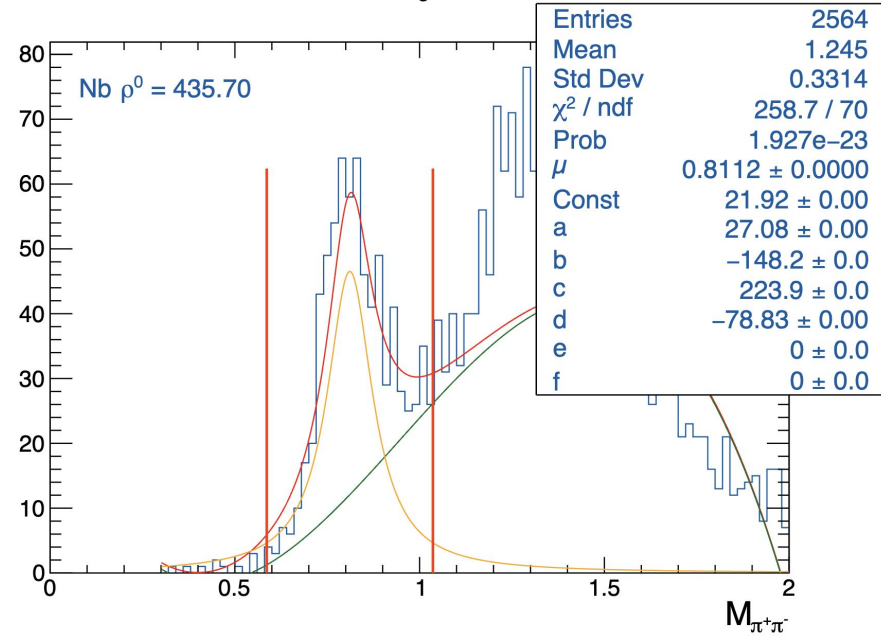
Background

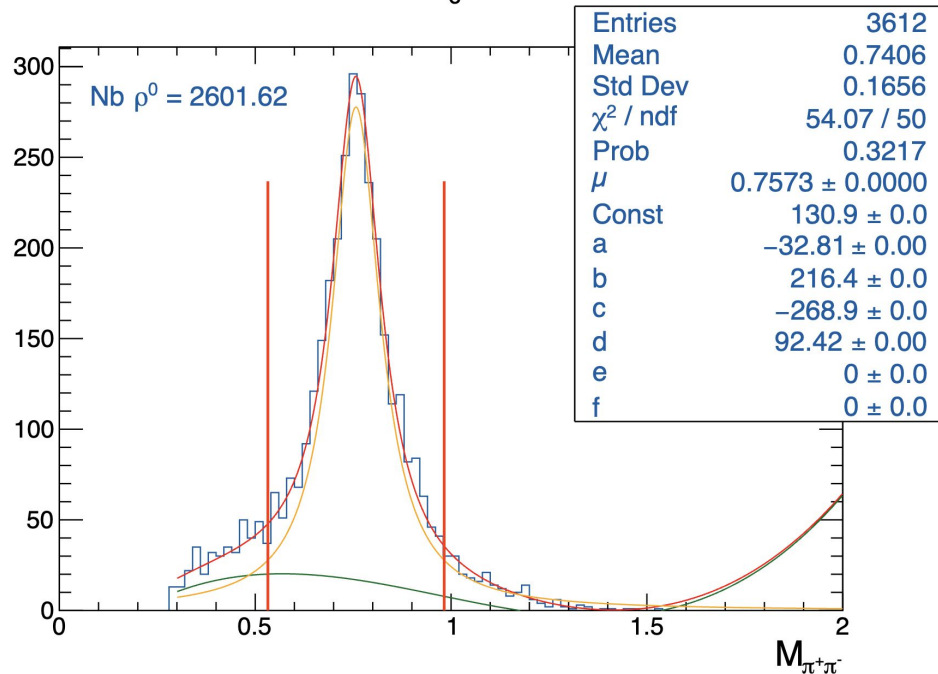
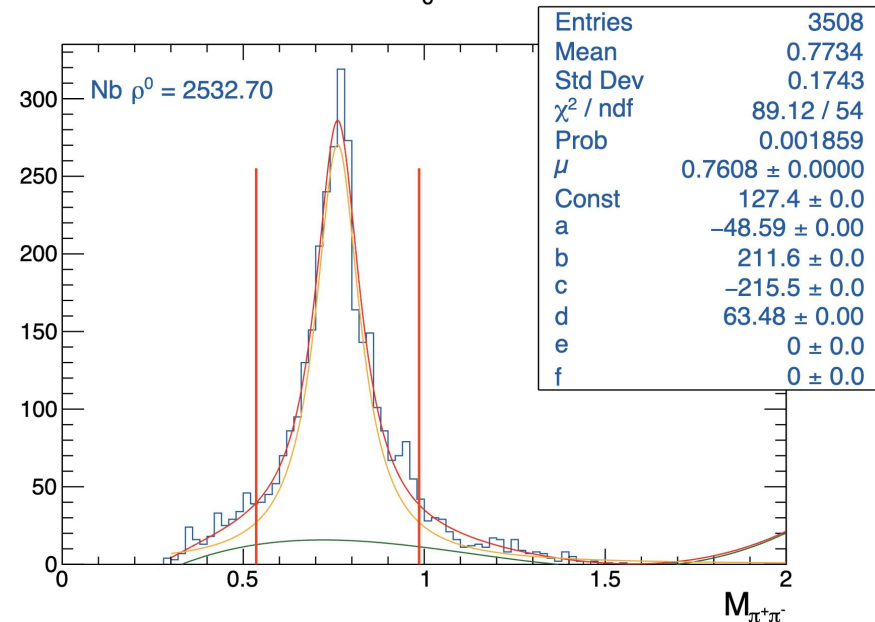
$0.65 < l_c < 0.73$  $0.73 < l_c < 1.00$ 

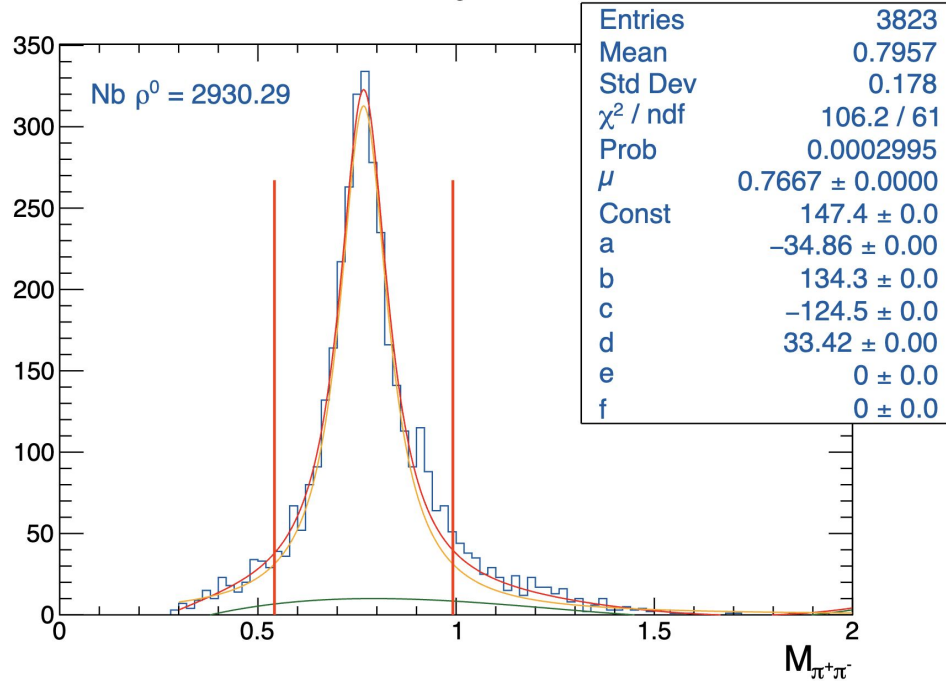
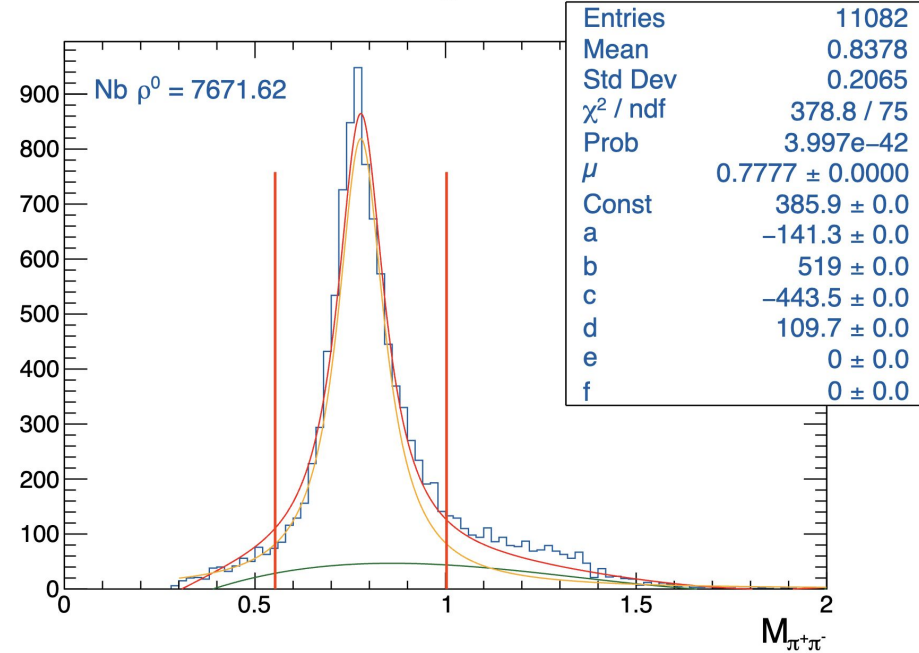
$1.00 < l_c < 2.00$  $2.00 < l_c < 3.00$ 

$0.00 < l_c < 0.57$

 $0.57 < l_c < 0.65$


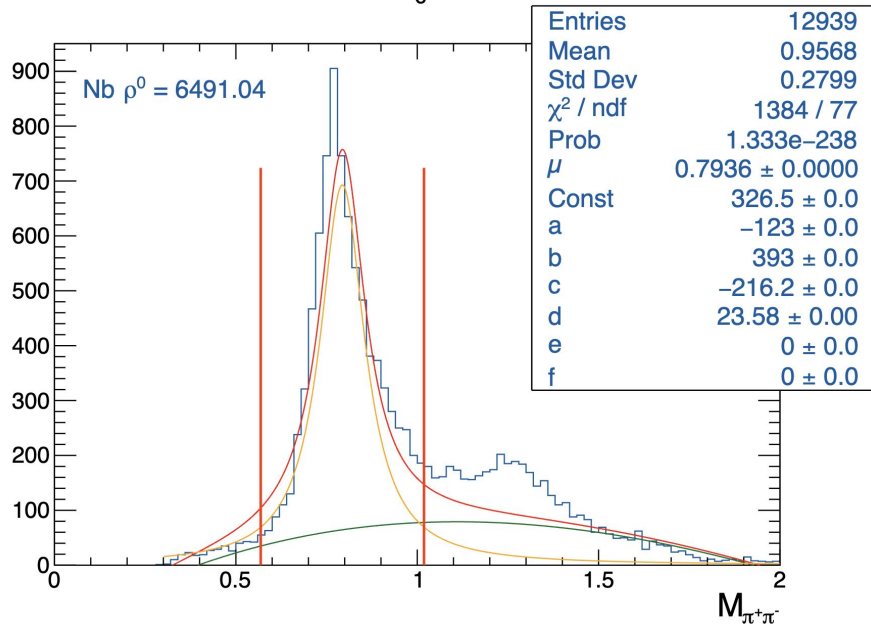
$0.65 < l_c < 0.73$  $0.73 < l_c < 1.00$ 

$1.00 < l_c < 2.00$  $2.00 < l_c < 3.00$ 

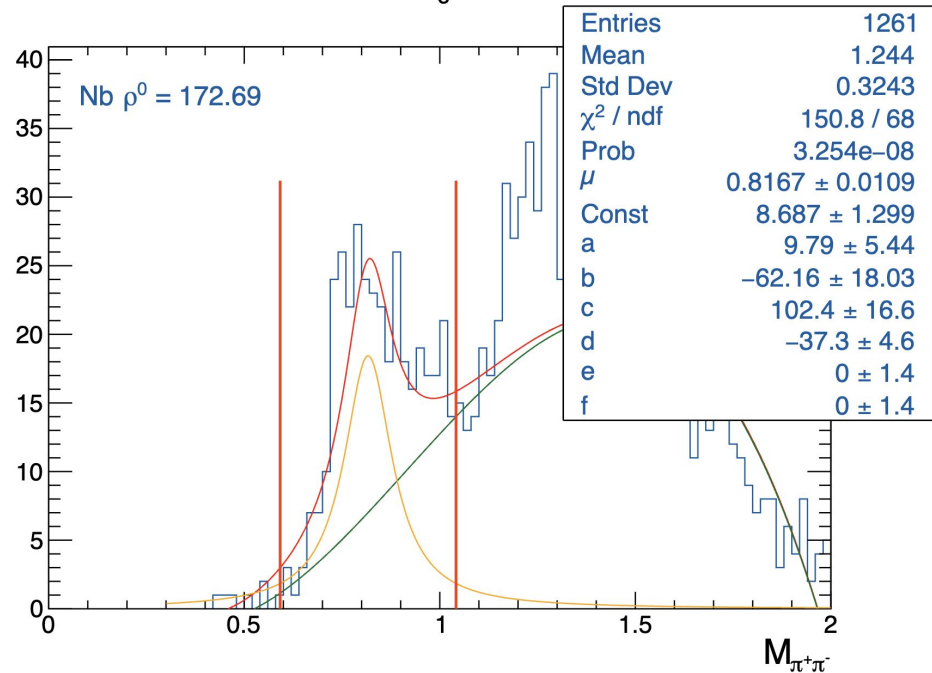
$0.00 < l_c < 0.57$  $0.57 < l_c < 0.65$ 

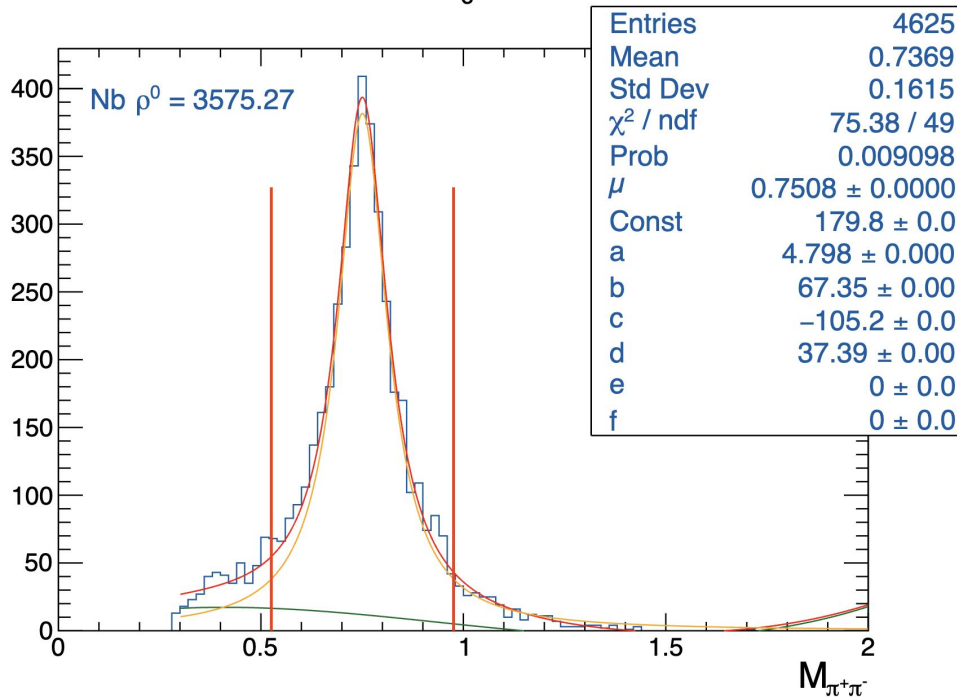
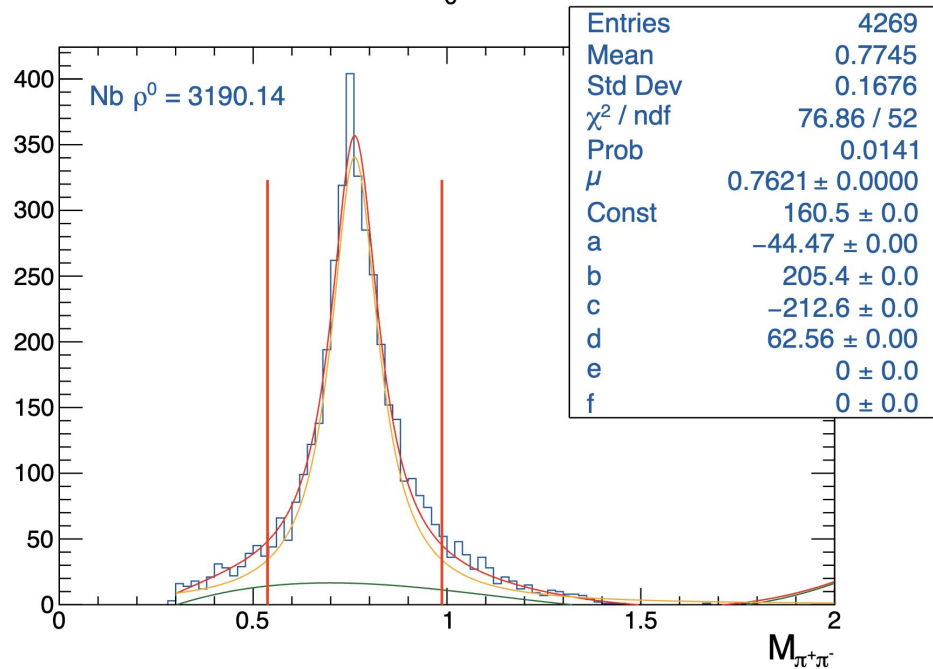
$0.65 < l_c < 0.73$

 $0.73 < l_c < 1.00$


$1.00 < l_c < 2.00$

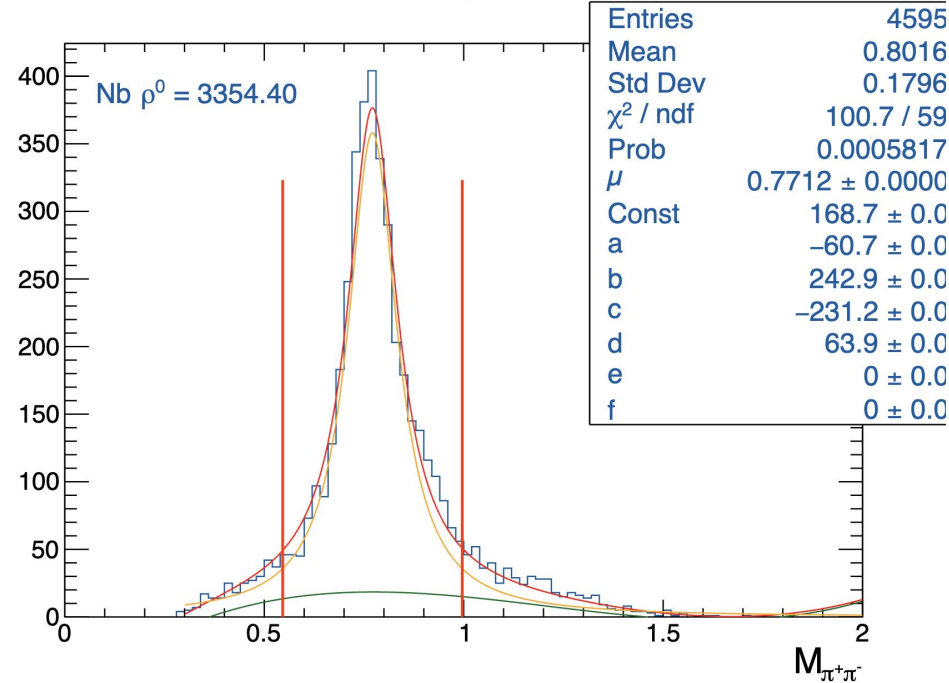


$2.00 < l_c < 3.00$

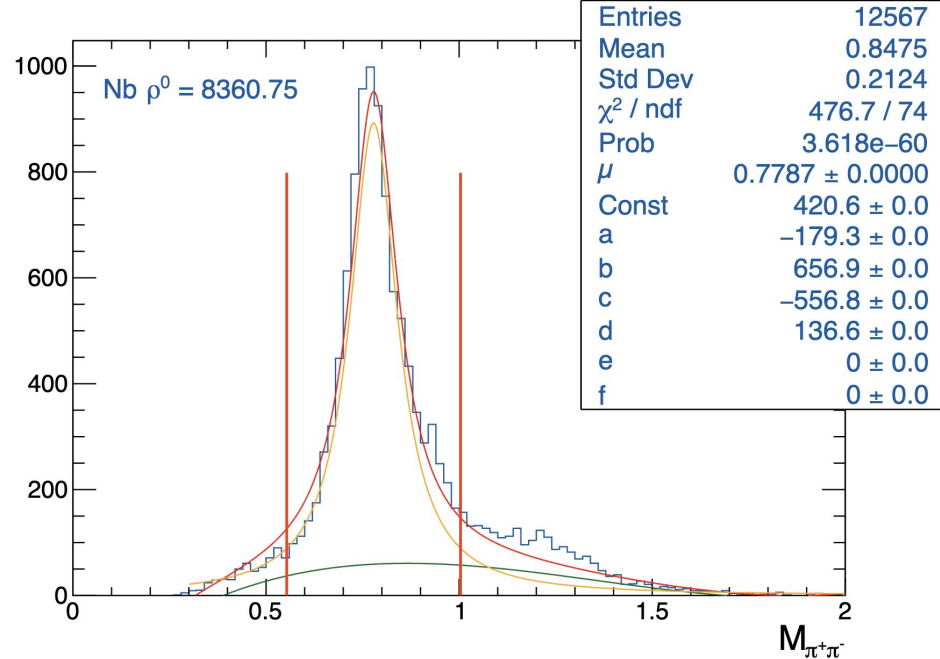


$0.00 < l_c < 0.57$  $0.57 < l_c < 0.65$ 

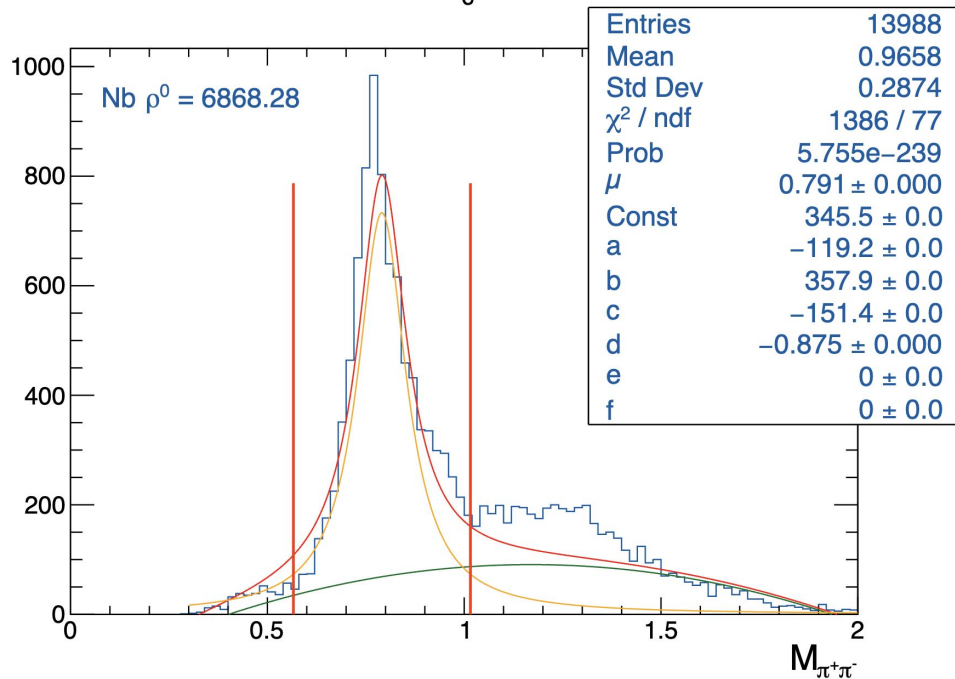
$0.65 < l_c < 0.73$



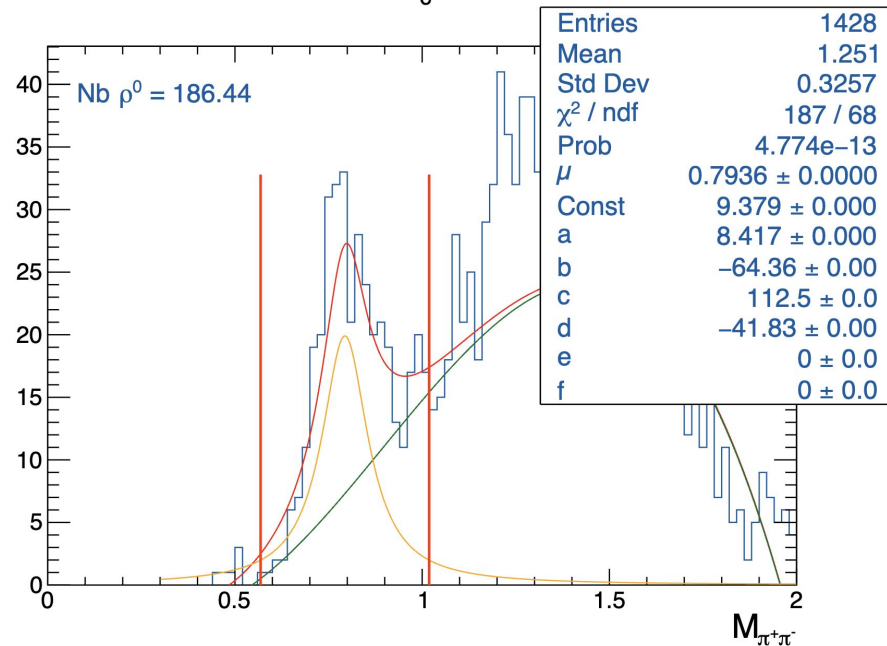
$0.73 < l_c < 1.00$



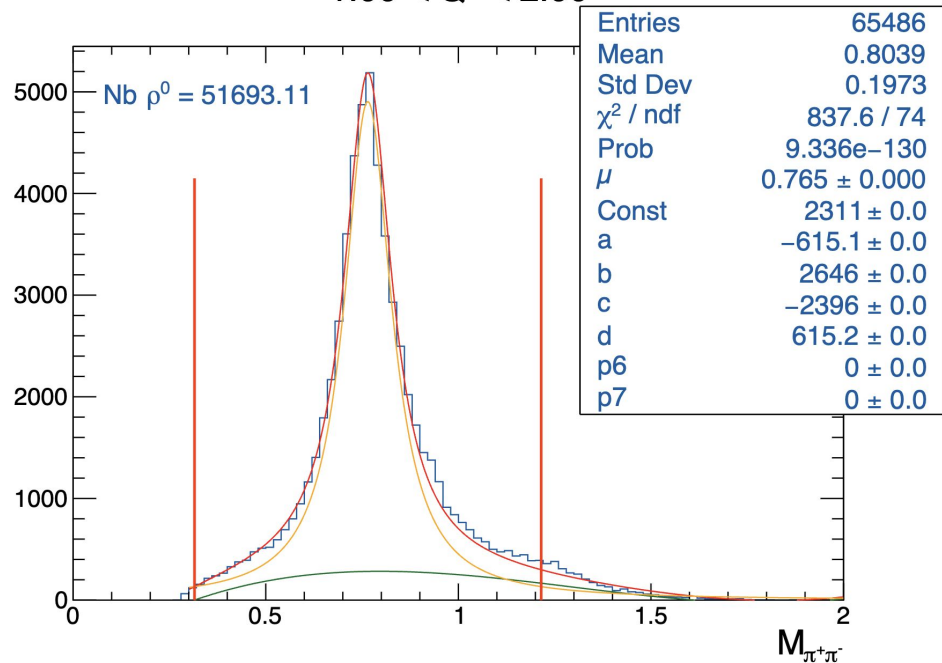
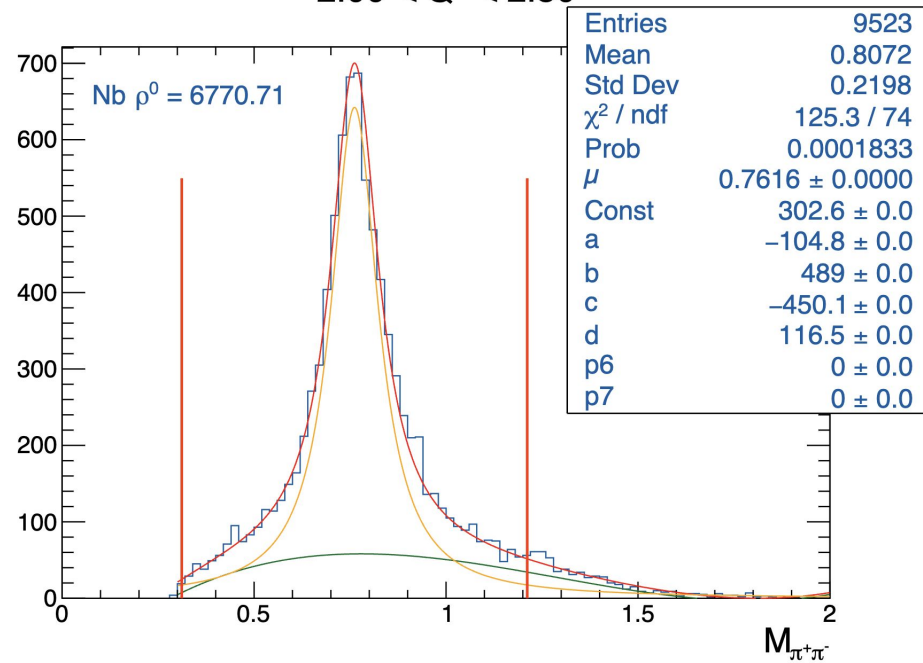
$1.00 < l_c < 2.00$

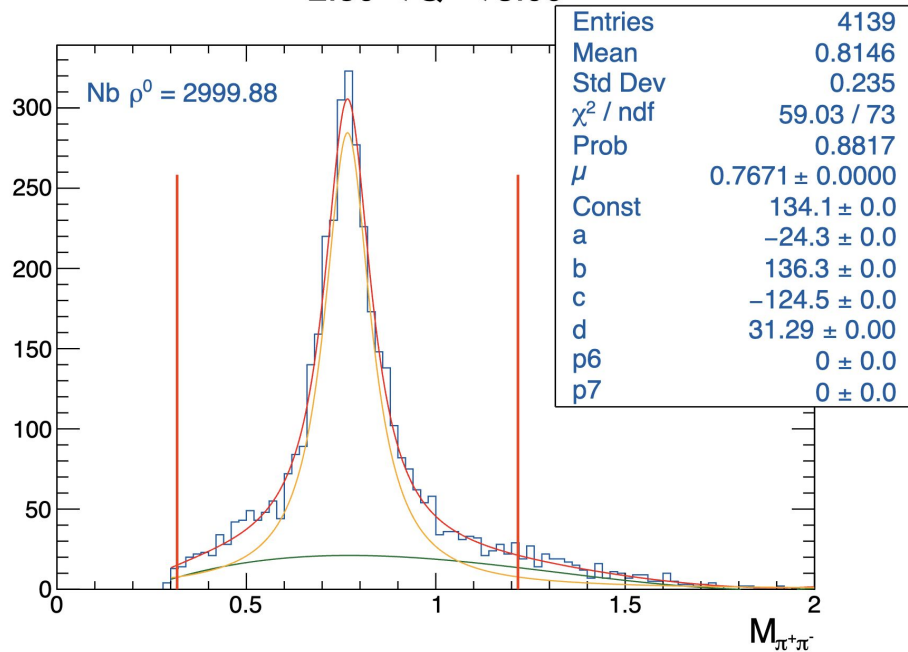
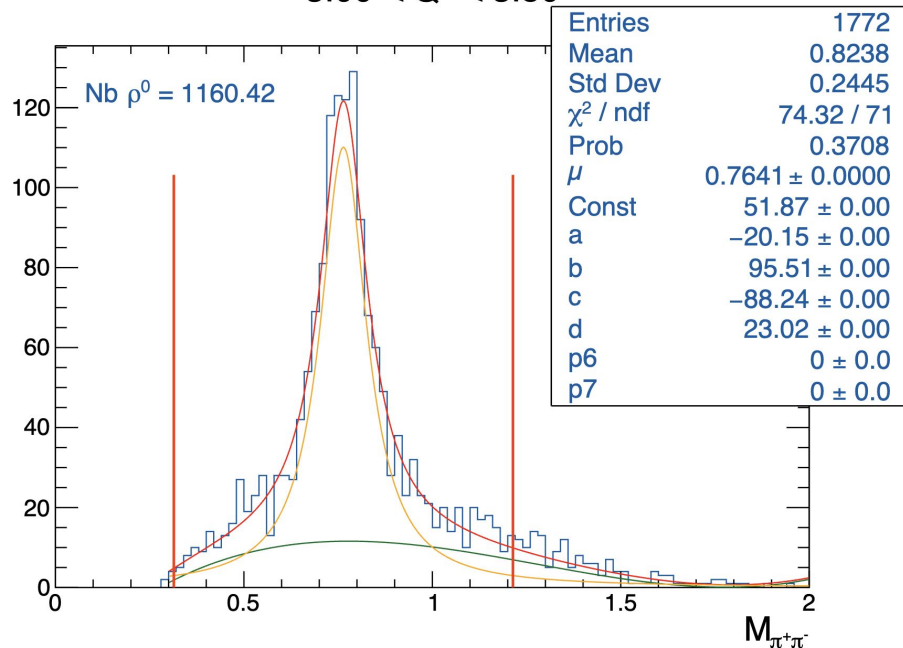


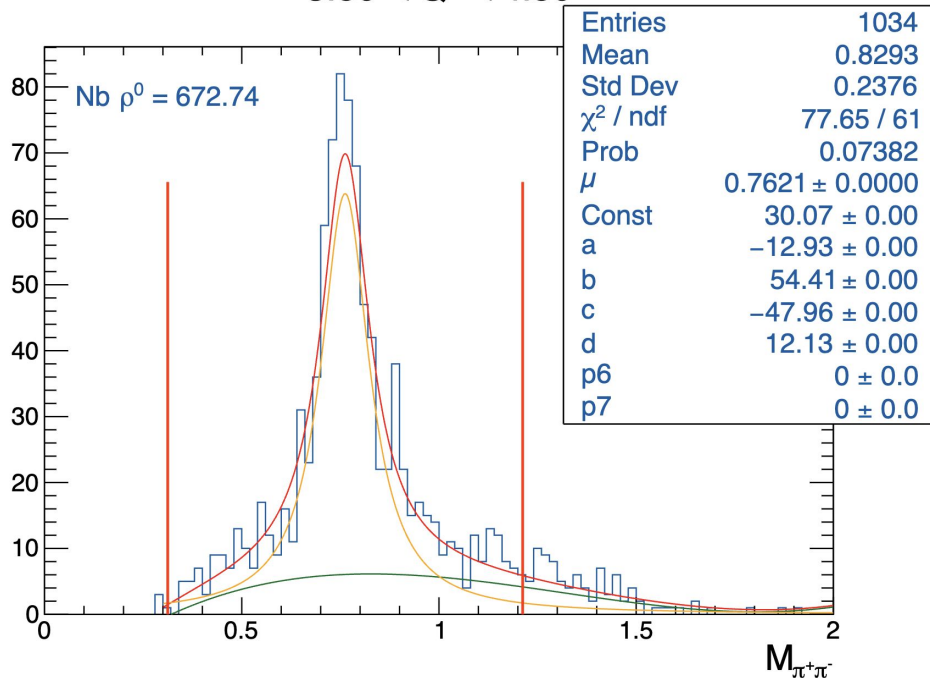
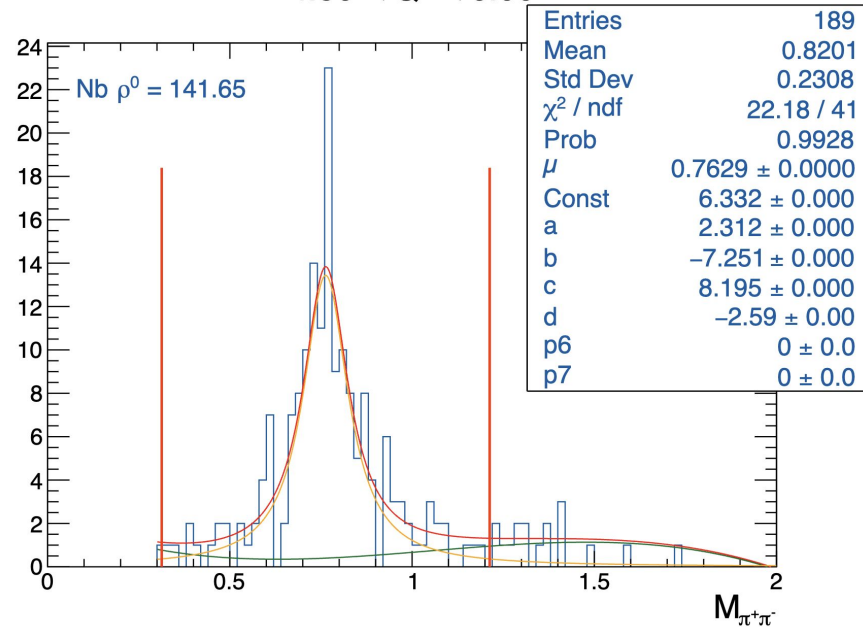
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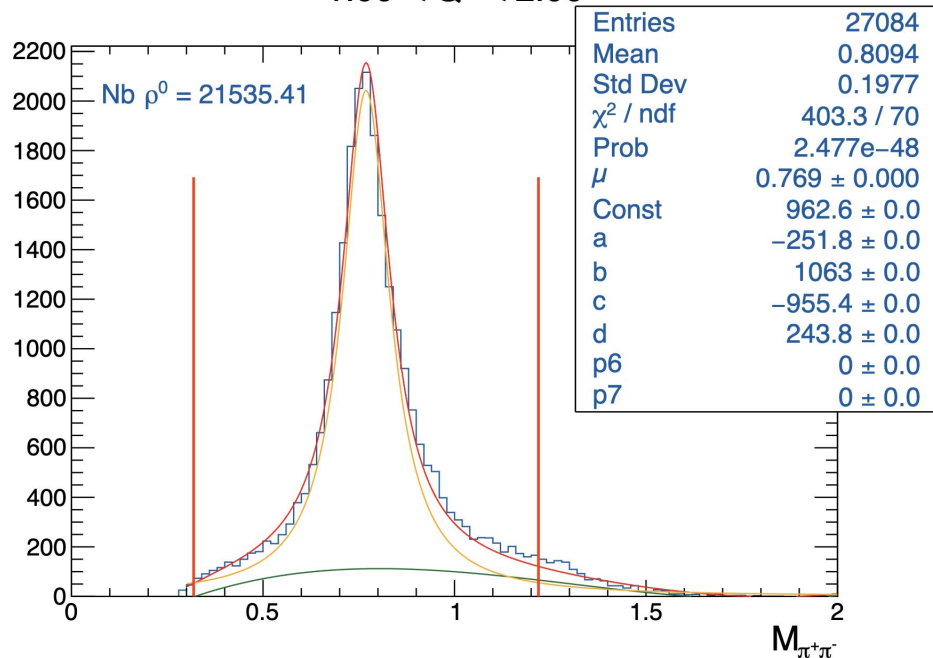
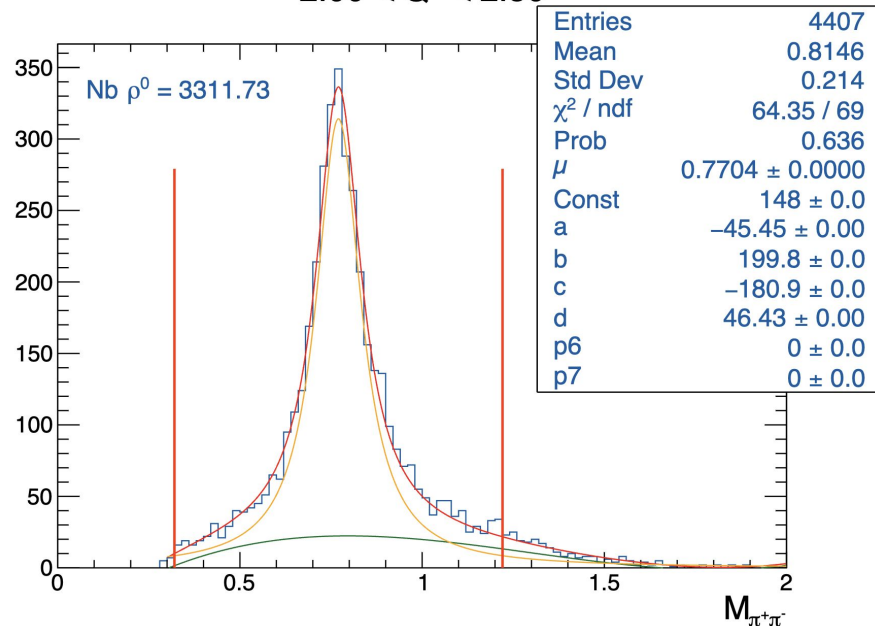


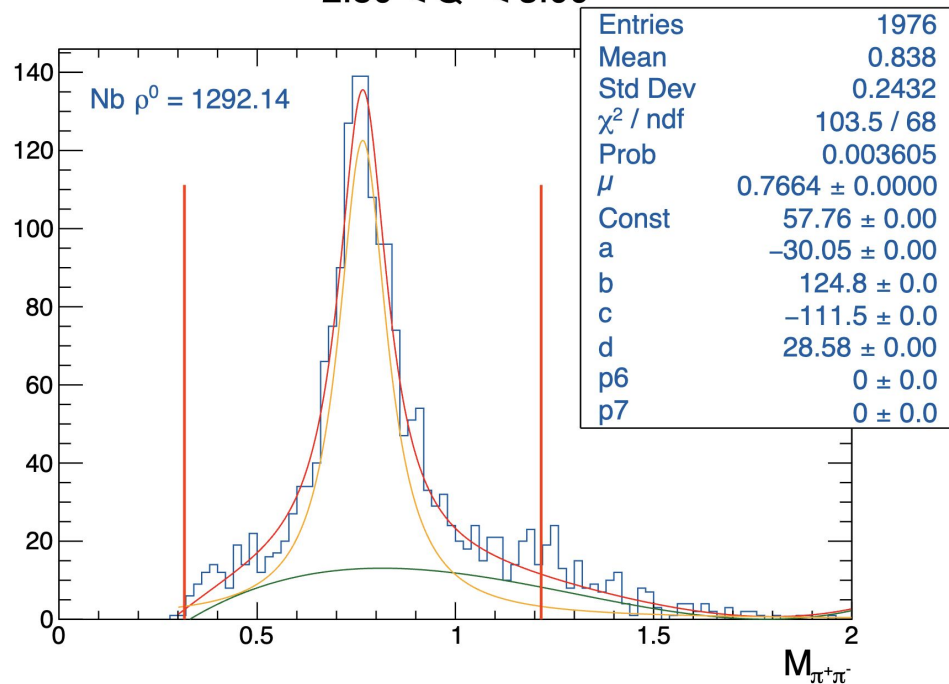
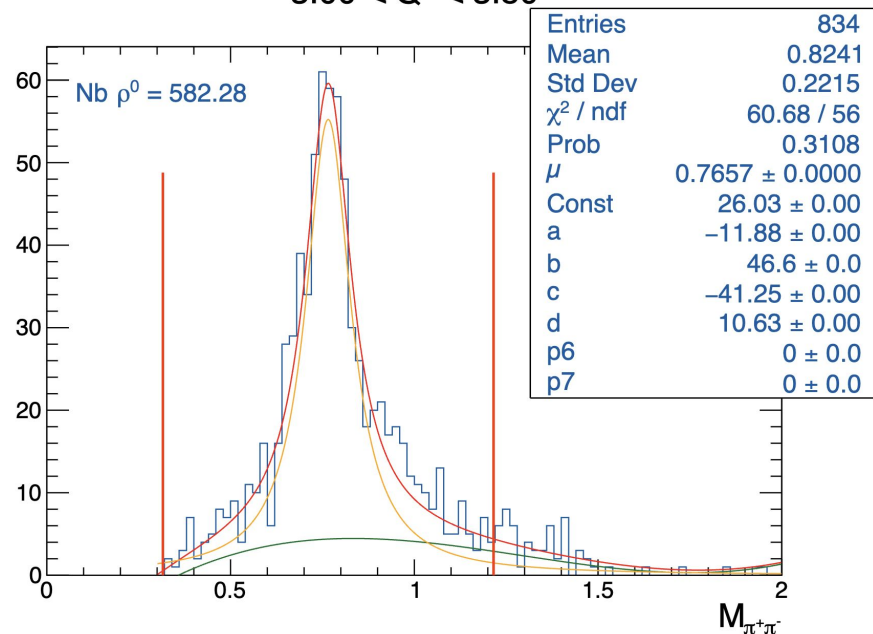
Nuclear transparency vs Q^2

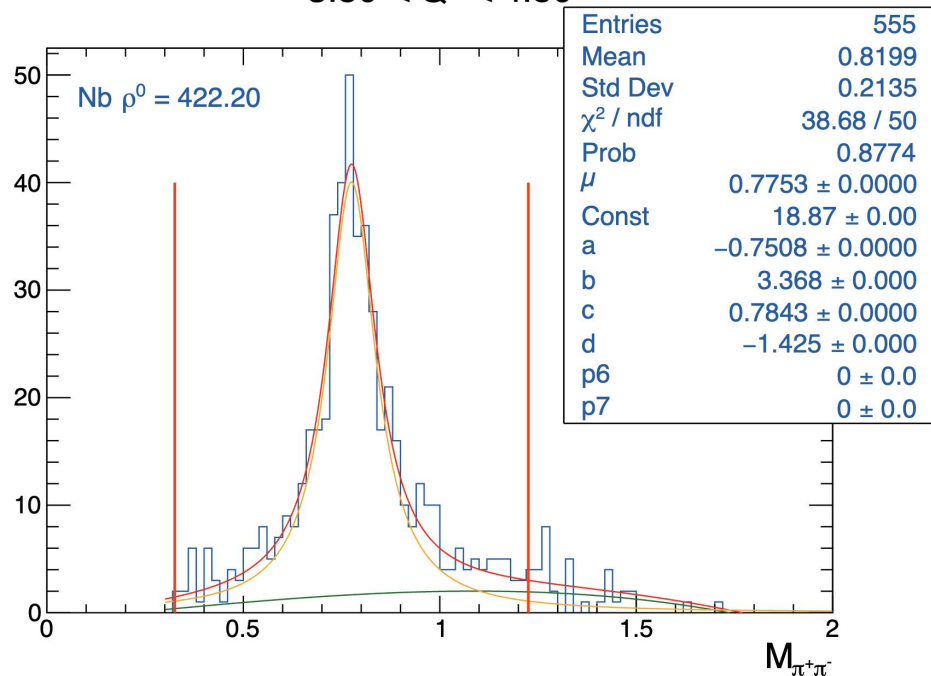
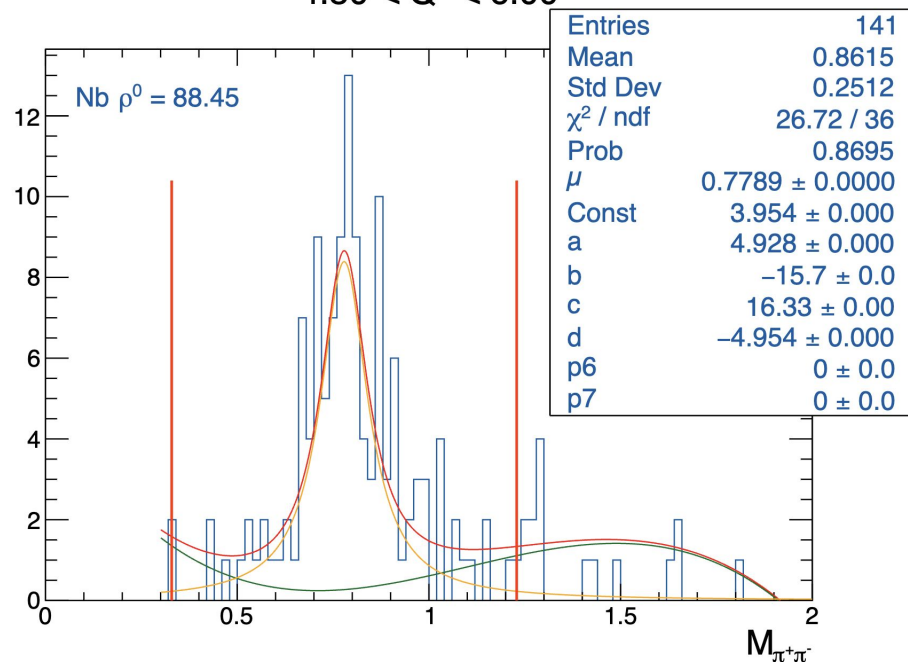
$1.00 < Q^2 < 2.00$  $2.00 < Q^2 < 2.50$ 

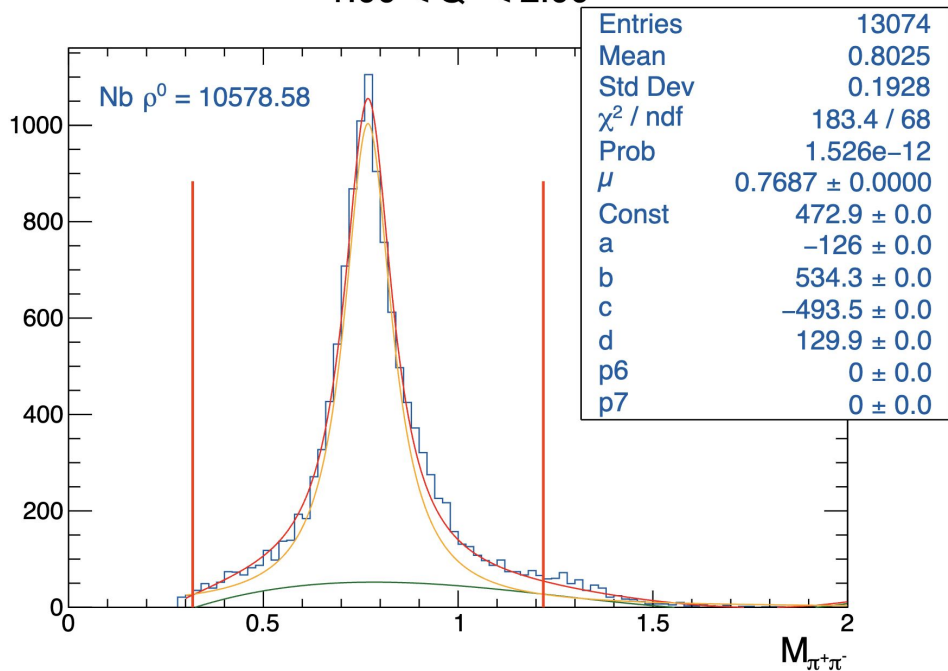
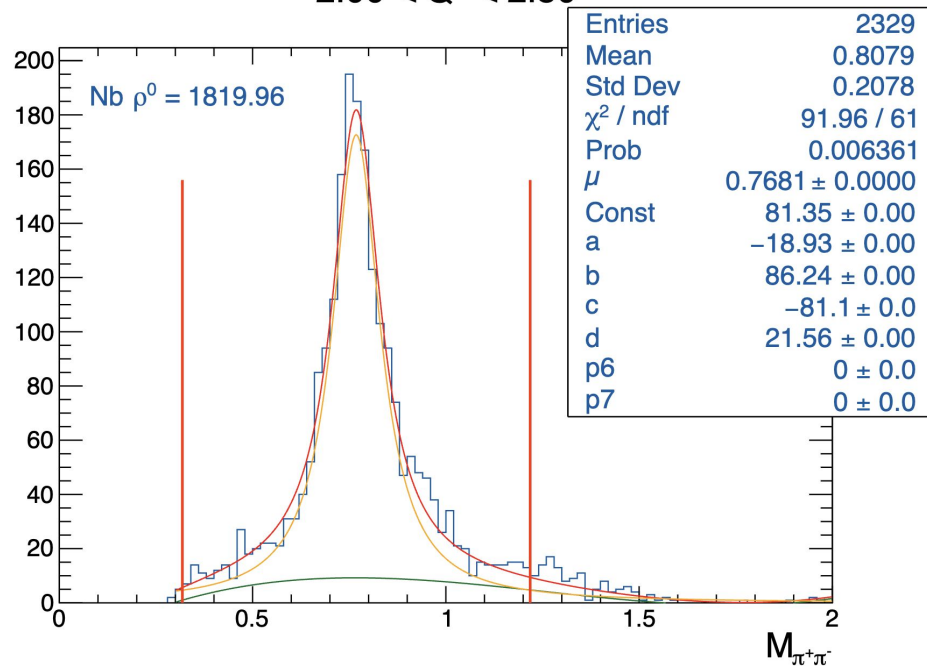
$2.50 < Q^2 < 3.00$  $3.00 < Q^2 < 3.50$ 

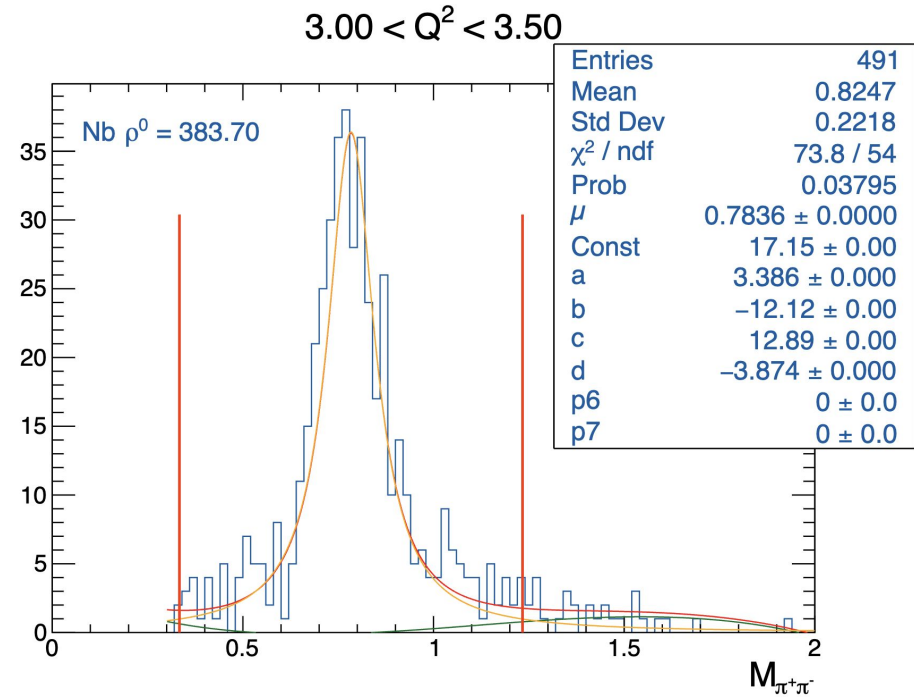
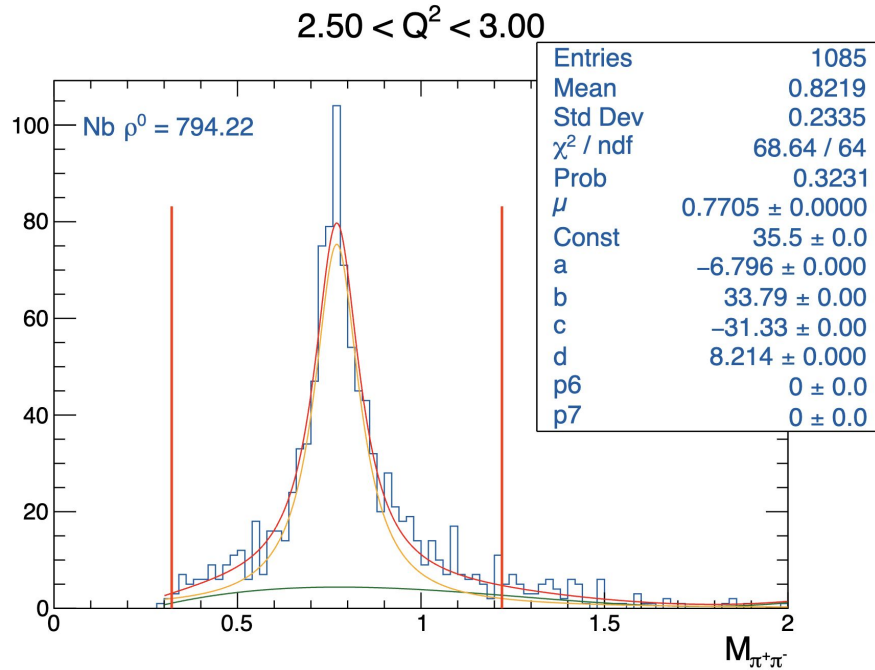
$3.50 < Q^2 < 4.50$

 $4.50 < Q^2 < 6.00$


$1.00 < Q^2 < 2.00$  $2.00 < Q^2 < 2.50$ 

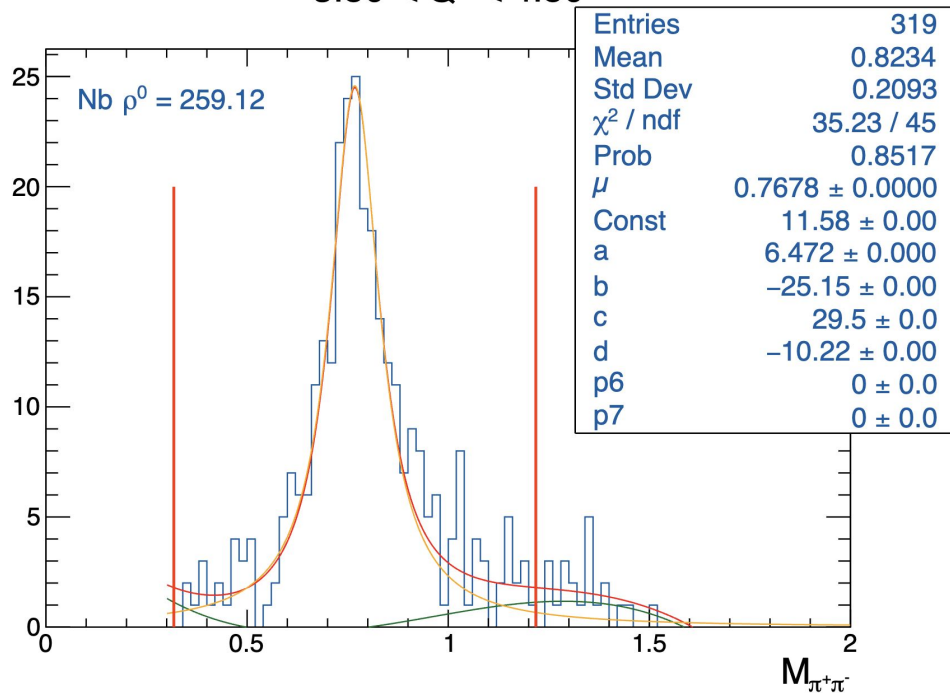
$2.50 < Q^2 < 3.00$  $3.00 < Q^2 < 3.50$ 

$3.50 < Q^2 < 4.50$  $4.50 < Q^2 < 6.00$ 

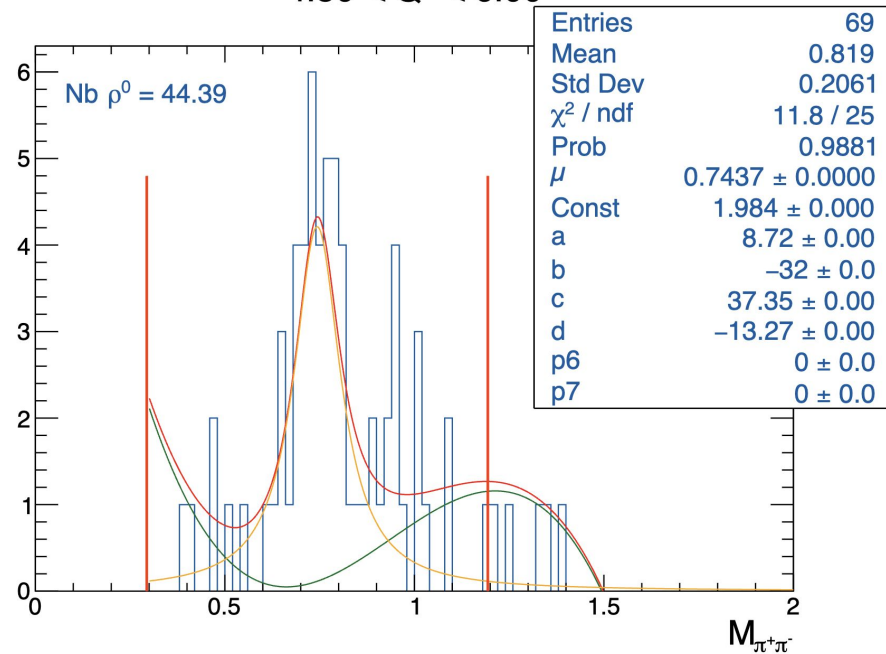
$1.00 < Q^2 < 2.00$

 $2.00 < Q^2 < 2.50$


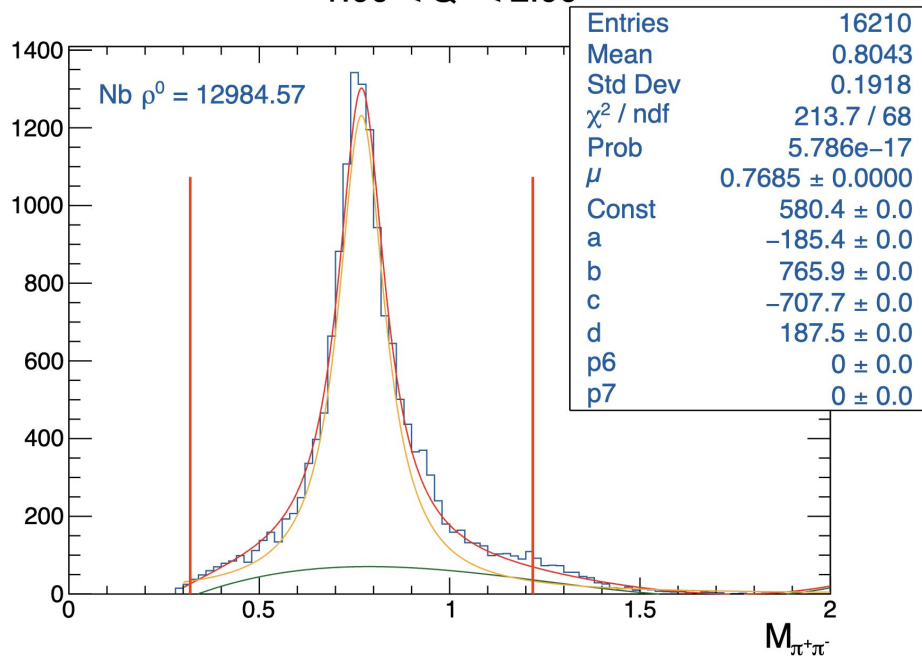
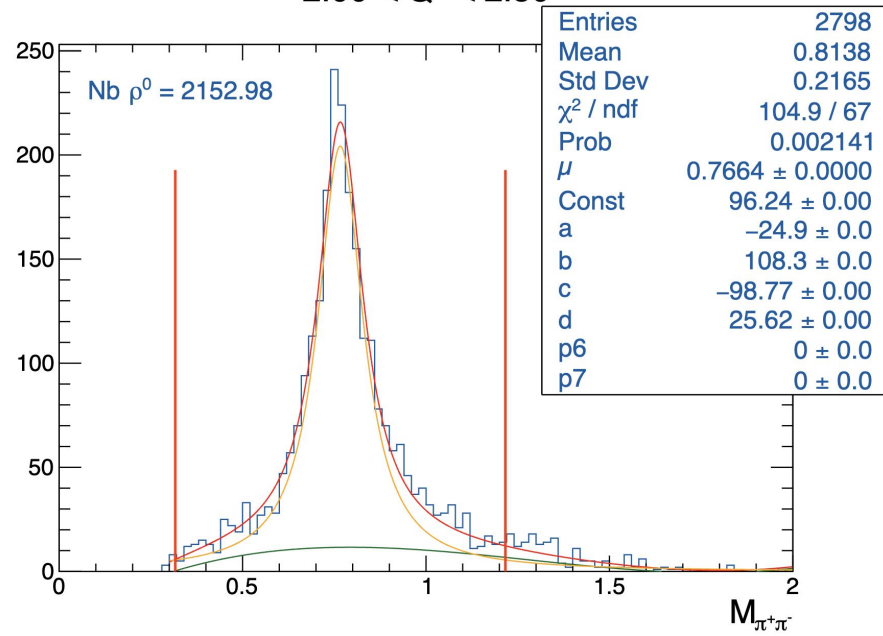


$3.50 < Q^2 < 4.50$

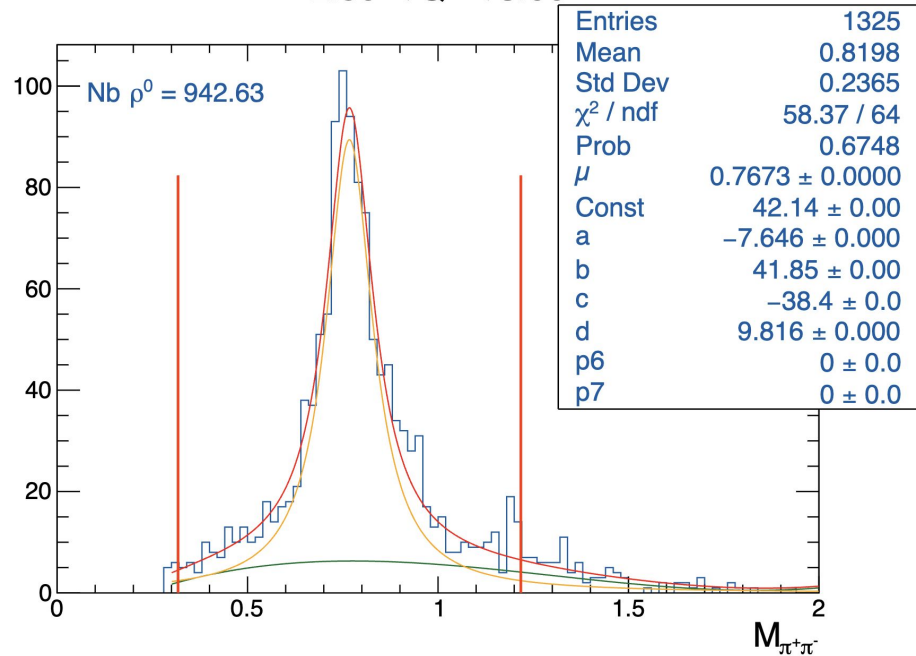


$4.50 < Q^2 < 6.00$

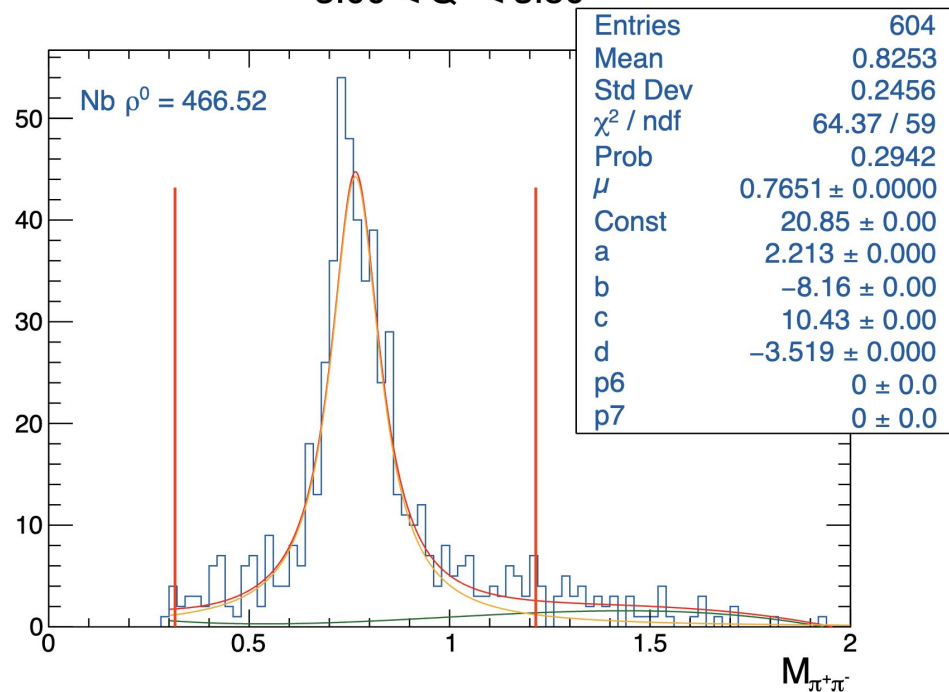


$1.00 < Q^2 < 2.00$

 $2.00 < Q^2 < 2.50$


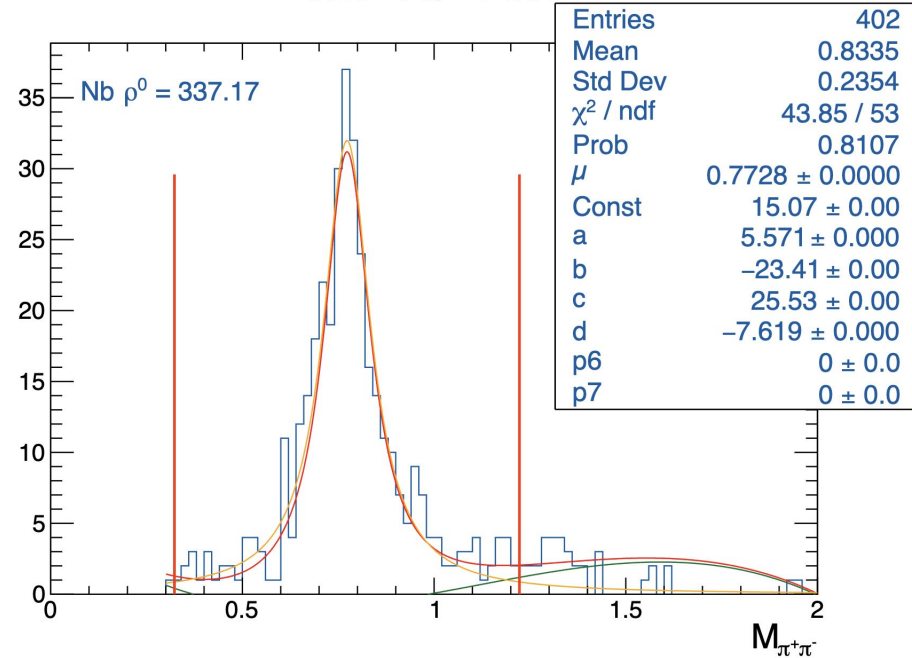
$2.50 < Q^2 < 3.00$



$3.00 < Q^2 < 3.50$



$3.50 < Q^2 < 4.50$



$4.50 < Q^2 < 6.00$

